

DO NOT WRITE ON TEST



**COMPUTER
PROGRAMMING
CONCEPTS
~OPEN EVENT~
(390)
NATIONAL 2022**

TOTAL POINTS _____ **100 Points**

Test Time: 60 minutes

MULTIPLE CHOICE

Identify the choice that *best* completes the statement or answers the question.

1. Packets that are transported across a network usually contain two elements of information: the control information and _____.
 - a. Encryption Keys
 - b. Delimiters
 - c. User Data
 - d. Addresses
2. What does a Java compiler create?
 - a. Executable
 - b. Bytecode
 - c. Machine code
 - d. Line by line instructions
3. Which term best describes the set of rules that defines the use of structures or expressions of a computer language?
 - a. Semantics
 - b. Syntax
 - c. Algorithm
 - d. Logic
4. Which term best describes a finite sequence of instructions to help a computer solve a problem?
 - a. Semantics
 - b. Syntax
 - c. Algorithm
 - d. Abstraction
5. What term best describes the type of language that is used to format the display of the text?
 - a. Interpreted
 - b. Compiled
 - c. Markup
 - d. High Level
6. What is a graphic tool that can help with planning and design of an algorithm?
 - a. Project Management Cycle
 - b. Business Proposal
 - c. Abstraction
 - d. Flow Chart

7. What can simulate a computer algorithm?
 - a. High level abstraction system
 - b. Flow chart
 - c. Turing-complete system
 - d. Prototype
8. What type of algorithm calls itself repeatedly until a certain condition is reached in which it will terminate?
 - a. Divide and conquer
 - b. Recursion
 - c. Logical
 - d. Serial
9. Which of the following is a characteristic of OOP?
 - a. Decomposing complex problems
 - b. Makes use of large sets of inference rules
 - c. Program executions is a systematic search for facts
 - d. Non mutable
10. What is another name for a front-end developer?
 - a. Database Programmer
 - b. Client Side Developer
 - c. Server Side Developer
 - d. DevOps Developer
11. Which of the following best describes a stack?
 - a. Last in first out data structures
 - b. Elements of the different data types being stored in the same data structure
 - c. Elements of the same data types being stored in the same data structure
 - d. Data structure that is used to temporarily place data as it is prepared for extraction
12. A subclass can have which of the following:
 - a. No methods inside of it
 - b. Proprietary created methods
 - c. Methods that have been redefined
 - d. All of these are correct
13. Which method will allow you to find out how many elements are contained in a static array?
 - a. size()
 - b. length
 - c. length ()
 - d. getSize()

14. Which term(s) would be the best alternative description for recursion?
- Iteration
 - OOP
 - Data abstraction
 - Inheritance
15. Compared to an ArrayList, a LinkedList provides better access to which of the following?
- First node
 - Last node
 - Middle node
 - First and last node
16. Which of the ratings below would indicate the best time for efficiency?
- $O(n)$
 - $O(\log n)$
 - $O(2^n)$
 - $O(n \log n)$
17. Based upon the efficiency ratings below, which one would be the *worst* choice for an algorithm?
- $O(2^n)$
 - $O(n)$
 - $O(n \log n)$
 - $O(\log n)$
18. Based upon the condition of locating a randomly selected element within an unordered array data structure, which of the following would represent the time complexity to find a specific value assuming you were using an efficient searching algorithm?
- $O(3)$
 - $O(n \log n)$
 - $O(n)$
 - $O(2^n)$
19. Based upon the condition of locating a randomly selected element within an ordered array data structure, which of the following would represent the time complexity to find a specific value assuming you were using an efficient searching algorithm?
- $O(n^2)$
 - $O(n \log n)$
 - $O(\log(n))$
 - $O(2^n)$
20. Based upon the condition of deleting the first element of a linked list, which of the following would represent the time complexity?
- $O(1)$
 - $O(3)$
 - $O(n \log n)$
 - $O(2^n)$

21. In order for an algorithm to perform a binary search in a complete binary search tree of n nodes, which of the following would represent the time complexity?
- $O(\log(n))$
 - $O(2 \log(n^2))$
 - $O(2 \log(n^3))$
 - $O(2 \log(n))$
22. When performing a linear search, which of the following would represent the worst-case complexity of efficiency?
- $O(2 \log(n^2))$
 - $O(n)$
 - $O(n \log n)$
 - $O(2^n)$
23. When performing a binary tree search, which of the following would represent the worst-case complexity of efficiency?
- $O(2 \log(n^2))$
 - $O(n)$
 - $O(n \log n)$
 - $O(2^n)$
24. The major disadvantages of the linked list and array are solved by which of the following?
- Binary trees
 - Stacks
 - Array list
 - Queues
25. Which of the following SQL statements selects the first, last and city columns from the "Employees" data table for everyone that is not from Seattle?
- SELECT first, last, city FROM Employees WHERE city < > S%
 - SELECT first, last, && city FROM Employees WHERE city < > Seattle
 - SELECT first, last, city FROM Employees WHERE city != Seattle
 - SELECT first, last, city FROM Employees WHERE city < > Seattle
26. Which of the following SQL statements selects the first, last, city and age columns from the "Employees" data table for everyone's last name ends in "re"?
- SELECT first, last, age FROM Employees WHERE where last LIKE '%re'
 - SELECT first, last, city, age FROM Employees WHERE where last LIKE '%re'
 - SELECT first, last, age FROM Employees WHERE where last LIKE '*re'
 - SELECT first, last, city, age FROM Employees WHERE where last LIKE '*re'

27. Which of the following SQL statements selects the first, last and age columns from the “Employees” data table for everyone’s first name is “Akaash”?
- SELECT first, last, age FROM Employees WHERE where first = ‘Akaash’
 - SELECT first, last, city, age FROM Employees WHERE where first = ‘Akaash’
 - SELECT first, last, city, age FROM Employees WHERE where first LIKE ‘Ak%’
 - SELECT first, last, city, age FROM Employees WHERE where first LIKE ‘%Ak%’
28. Which of the following SQL statements creates a table called “Employees” with information for the first name, last name and age; the names have a max character input of 30 and the age has a max number of digits set at 2.
- CREATE TABLE Employees (FirstName maxchar(30), LastName maxchar(30), Age maxchar (2));
 - INSERT TABLE Employees (FirstName varchar(30), LastName varchar(30), Age number (2));
 - CREATE SELECT TABLE Employees (INSERT FirstName varchar(30), INSERT LastName varchar(30), INSERT Age number (2));
 - CREATE TABLE Employees (FirstName varchar(30), LastName varchar(30), Age number (2));
29. Which of the following SQL statements correctly inserts data into a table?
- INSERT into Employee (first, last, age, address, city, state) values (‘Amber’, ‘Rodriguez’, 23, ‘234 Haskell Ave’, ‘Atlanta’, “Georgia”);
 - CREATE into Employee (first, last, age, address, city, state) values (‘Amber’, ‘Rodriguez’, 23, ‘234 Haskell Ave’, ‘Atlanta’, “Georgia”);
 - SELECT Employee INSERT into (first, last, age, address, city, state) values (‘Amber’, ‘Rodriguez’, 23, ‘234 Haskell Ave’, ‘Atlanta’, “Georgia”);
 - INSERT into Employee CREATE into (first, last, age, address, city, state) values (‘Amber’, ‘Rodriguez’, 23, ‘234 Haskell Ave’, ‘Atlanta’, “Georgia”);
30. What shape in a flowchart known as an off-page connector and is used to connect to targets that are on another page?
- Open triangle with a dashed or solid line
 - Rectangle with double struck vertical edges
 - Pentagon
 - Circle
31. What shape in a flowchart known as manual operation and is used to indicate an operation that has to be made manually?
- Cylinder/disk drive
 - Trapezoid (longest parallel side at the top)
 - Pentagon
 - Quadrilateral (sloping up from left to right)

32. What shape in a flowchart is known as a document in a program?
- Open rectangle with a dashed or solid line
 - Rectangle with double struck vertical edges
 - Pentagon
 - Rectangles with a wavy base
33. What shape in a flowchart known as manual input?
- Cylinder/disk drive
 - Hexagon
 - Pentagon
 - Quadrilateral (sloping up from left to right)
34. What shape in a flowchart known as a preparation or initialization is used for steps in a program like setting a switch or initializing a sub routine?
- Rectangle
 - Diamond/Rhombus
 - Hexagon
 - Trapezoid (longest parallel side at the top)
35. Which term best describes a language that allows the developers to create a way to visualize a systems architectural blueprints including activities, components and its systems?
- UML
 - OOP
 - CASE
 - RAD
36. What is a benefit of ULM?
- Demonstrates how components and interfaces interact with one another
 - Visualizes how a system will run
 - Helps programmers understand associated jobs for the program
 - All of these are benefits
37. What type of view in a ULM model shows a structural perspective of the objects, attributes, and operations?
- Dynamic
 - Static
 - Activity
 - State
38. What is a drawback to the use of ULM?
- Structure programs only emphasize the things that must be present in the model
 - Sequence models reveals communication between objects
 - Multiple Meta Object Facility layers
 - It is overused causing needless details

39. What value is stored by the statement: `int d = ((int)-91.1 % 3)`

- a. -1
- b. -1.0
- c. 1
- d. 1.0

40. What value is stored by the statement: `int d = ((int)-91.1 % 3 % (int)3.0)`

- a. -1
- b. -1.0
- c. 1
- d. 1.0

41. What will be displayed by the method call `testRun ('A', ('z'/2))`?

```
public static void testRun(int x, int y)
{
    if (x <= y) {
        print(x + " ");
        testRun (x+1, y-1);
        print(y + " ");
    }
}
```

- a. 65 66 67 68 68 69 70 71 72 73 74 75
- b. 65 66 67 68 68 69 70 71 72 73 74
- c. 65 66 67 68 68 69 70 71 72 73
- d. Nothing will display

42. What will be displayed by the method call `testRun (10,5)`?

```
public static int testRun(int x, int y)
{
    if (x < y) {
        print(" "+ x);
        return x;
    }else {
        print(" "+ x);
        return y + testRun(x-1,y+1);
    }
}
```

- a. 10 9 8 7 6 5
- b. 10 9 8 7 6
- c. 10 9 8 7
- d. 10 9 8

43. What will be displayed by the method call *testRun (379,345)*?

```
public static int testRun(int x, int y)
{
    if (x % y == 0) {
        print(y + " ");
        return y;
    }
    else {
        print(x + " ");
        return testRun(y, x % y);
    }
}
```

- a. 379 345 34 5 1 0
- b. 379 345 34 10 1
- c. 379 345 34 5 1
- d. Nothing will print

44. What will be displayed by the method call *testRun (2)*?

```
public static int testRun(int x)
{
    if (x == 0) {
        print(x + " ");
        return 1;
    }
    else {
        print(x + " ");
        return x + testRun(x-1) + testRun(x-1) + testRun(x-1);
    }
}
```

- a. 2 1 0 0 0 1 0 0 0 1 0 0 0
- b. 2 1 0 0 0 1 0 0 0 1 0 0
- c. 2 1 1 0 0 1 0 0 0 1 0 0 0
- d. Nothing will print

45. What will be displayed by the program segment below?

```
int twoD [ ][ ];
twoD = new int[3][4];
int size = twoD.length;
double fill = 3.0;
for(int a = 0; a < size; a++)
{
    for(int b = a; b < size; b++)
    {
        twoD[a][b] = (int)(fill*3/2);
        print(" " + twoD[a][b]+ " ");
        fill+=fill;
    }
    print("\n");
}
println();
```

- a. 4 9 18 || 36 72 ||
- b. 4 9 18 || 36 72 || 144 ||
- c. 4 9 18 36 || 72 144 288 || 576 1152 ||
- d. ArrayIndexOutOfBoundsException

46. What will be displayed by the program segment below?

```
int twoD [ ][ ];
twoD = new int[3][4];
int size = twoD.length;
double fill = 3.0;
for(int b = 0; b < size; b++)
{
    for(int a = b; a < size; a++)
    {
        twoD[a][b] = (int)(fill*3/2);
        print(" " + twoD[a][b]+ " ");
        fill+=fill;
    }
    print("\n");
}
println();
```

- a. 4 9 18 || 36 72 ||
- b. 4 9 18 || 36 72 || 144 ||
- c. 4 9 18 36 || 72 144 288 || 576 1152 || 2304 ||
- d. ArrayIndexOutOfBoundsException

47. What is printed by the statement:

```
char b = 65;  
b += 65;  
if (b > 'z')  
    print("This is " +b);  
else  
    print("It really should be " +b);
```

- a. This is ?
- b. This is 130
- c. It really should be ?
- d. It really should be 130

48. What is printed by the statement:

```
int a = (char)(165.5 %100);  
char b = 65;  
b += a + b;  
int c = (int)b/3;  
  
if (c > (a+b))  
    print("This is " +c);  
else  
    print("It really should be " +c);
```

- a. This is 65
- b. This is A
- c. It really should be 65
- d. It really should be A

49. When the program finishes running, what will be printed in the console?

```
String div = "cat";  
int x = 233;  
  
switch(div)  
{  
    case "cat": x += 1;  
    case "dog": x += 2;  
    case "bird": x += 3;  
    case "cow": x += 4;  
}  
print(x);
```

- a. 233
- b. 234
- c. 243
- d. Nothing will print

50. When the program finishes running, what will be printed in the console?

```
char div = 'A';  
int x = 130/div;  
  
switch(div)  
{  
    case 'A': x += div;  
    case 'B': x += div + 2;  
    case 'C': x += div + 3;  
    case 'D': x += div + 4;  
}  
print(x);
```

- a. 65
- b. 134
- c. 271
- d. 202